Computational Problem Spaces and Methods for Truck and Resource Orchestration in Logistics Terminals: A Systematic Literature Review

Marcus Vinicius[a,*]

[a] *PequiFlux Research Group, Brazil*

Abstract

This paper presents a systematic literature review of computational problem spaces and solution approaches for truck and resource orchestration in logistics terminals. The review examines how the literature models recurrent operational pain points, including gate congestion, truck-resource synchronization, internal balancing, and disruption-driven rescheduling, and how these problems are translated into optimization, simulation, and hybrid decision artifacts. Following a Parsifal- and PRISMA 2020-based protocol, searches in IEEE Xplore, Web of Science, and Scopus retrieved 60 records; after screening, backward snowballing, quality assessment, and a terminal-focused scope refinement, 31 studies were retained in the final synthesis. The evidence base is dominated by mathematical programming, metaheuristics, and simulation, with strong concentration in cross-docking and container-terminal settings. Across the corpus, most studies remain focused on temporally bounded coordination problems, whereas subsystem coupling, richer uncertainty representations, and continuously adaptive control are less common. Validation is treated here as an analytical lens of operational maturity rather than as the sole object of the review: many methods perform well in solver-based or simulation-based settings, but only a small share of the literature provides stronger operational grounding. The review therefore positions the field as methodologically rich but still uneven in deployment-oriented evidence, and highlights integrated terminal models, richer disturbance representations, shared benchmarks, and more operationally grounded testbeds

*Corresponding author

Email address: marcux7777@gmail.com (Marcus Vinicius)

as central directions for future research.

Keywords: truck scheduling, cross-docking, logistics terminals, systematic literature review, container terminals

---

1. Introduction

Truck and physical resource orchestration has become a critical decision problem in logistics terminals. Gates, docks, yards, transfer equipment, and truck arrivals interact under severe capacity constraints, and poor synchronization increases waiting times, tardiness, and operational cost. In container terminals and cross-docking systems, this challenge is sharper because arrivals are dynamic, resources are heterogeneous, and decisions often need continuous revision [1–3].

The literature has responded with mathematical programming for terminal scheduling and allocation [4–7], simulation and agent-based control [8, 9], and, more recently, hybrid data-driven and learning-based strategies [10–14]. Here, the focus is on adaptive and data-coupled orchestration, especially pipelines that combine predictive or event-stream-driven signals with optimization or simulation.

Even with this methodological diversity, the field remains fragmented. Many studies optimize only one subsystem at a time, and uncertainty is often restricted to arrival times or service durations, with fewer studies addressing richer disruptions such as resource failures, no-shows, or inconsistent real-time data. Several authors explicitly note a persistent gap between academic formulations and industrial needs, especially when static or offline assumptions are used to represent dynamic operations [4, 12, 15, 16]. This paper reads that gap through a *validity frontier*: the boundary between algorithmic success in stylized models and evidence that remains credible under operational complexity.

Adjacent review efforts have mostly focused on narrower slices of the problem, such as cross-docking scheduling, truck appointment systems, or integrated scheduling in automated terminals [17–19]. This paper uses that validity frontier as its review lens and examines how far terminal-orchestration research has progressed toward operationally grounded, data-coupled orchestration.

2. Review Method

The review protocol was designed in Parsifal and reported according to PRISMA 2020 guidelines [20]. Figure 1 summarizes study selection. Planning followed a PICOC structure centered on terminal truck scheduling, computational methods, and outcomes such as queue, congestion, cost, and makespan reduction.

The review was guided by three research questions:

1. Which computational approaches have been proposed for truck scheduling and physical resource allocation in high-traffic logistics terminals?
2. How do the studies model objectives, constraints, dynamics, uncertainty, and technical artifacts?
3. Which limitations remain regarding real-time operation, disruption-aware re-optimization, and practical applicability?

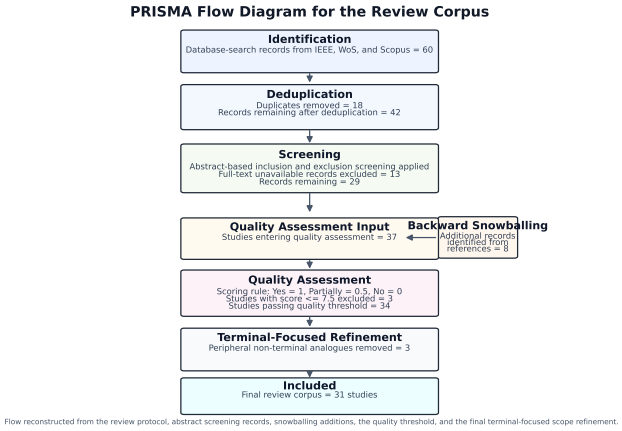

Figure 1: Study selection flow from database retrieval, screening, backward snowballing, quality assessment, and terminal-focused scope refinement.

2.1. Search Strategy and Eligibility

The search strategy combined terminal-context, method, and performance terms. All database searches in IEEE Xplore, Web of Science, and Scopus were conducted on February 27, 2026, followed by backward snowballing through Zotero in the same review cycle. A preserved Parsifal export in the working archive lists source-level imports of 12 records from IEEE Digital Library, 20 from Web of Science, and 22 from Scopus, plus 8 additional

records identified via Zotero-supported backward snowballing. Because these source-level import counts were preserved in a separate working snapshot, the PRISMA flow reported below remains the auditable basis for the screening and retention totals used in this manuscript. The preserved Boolean core of the protocol was:

> ("agribusiness logistics" OR "cross-docking" OR "dock doors" OR "gates" OR "terminal logistics" OR "truck appointment system" OR "truck arrivals" OR "yard management" OR "yard trucks")
> AND
> ("dynamic scheduling" OR "heuristic" OR "machine learning" OR "optimization" OR "predictive-reactive scheduling" OR "real-time scheduling" OR "reinforcement learning" OR "resource allocation" OR "simulation")
> AND
> ("bottleneck reduction" OR "congestion reduction" OR "delay reduction" OR "dock utilization improvement" OR "makespan reduction" OR "operational cost reduction" OR "queue reduction" OR "waiting time reduction")

Table 1 records the source-level execution details that remain recoverable from the current workspace and companion materials.

The common Boolean core above remained the conceptual search backbone across sources. For IEEE Xplore, the exact archived query was:

```
( "truck scheduling" OR "truck appointment system" OR "yard management"
OR "terminal logistics" OR "agribusiness logistics" )
AND ( queue* OR "resource allocation" OR bottleneck OR dock* )
AND ( optimiz* OR heuristic* OR "reinforcement learning" OR "machine
learning" OR simulation )
AND ( "real-time" OR dynamic OR uncertain* )
```

IEEE Xplore showed 13 results for this query, while the preserved Parsifal import snapshot lists 12 IEEE records. The retrospective interface hit count is therefore informative for reproducibility, but the archived screening flow and PRISMA totals remain anchored to the preserved import and screening records. For Web of Science, the exact archived advanced-search syntax was:

```
TS=(( "truck scheduling" OR "truck appointment system" OR "yard management"
OR "terminal logistics" OR "agribusiness logistics" )
AND ( queue* OR "resource allocation" OR bottleneck OR dock* )
AND ( optimiz* OR heuristic* OR "reinforcement learning" OR "machine
learning" OR simulation )
AND ( "real-time" OR dynamic OR uncertain* ))
```

With the publication restriction 2021–2026 applied, Web of Science returned 21 results, while the preserved Parsifal import snapshot lists 20 Web of Science records. As with IEEE Xplore, the retrospective interface hit count improves reproducibility, but the archived screening flow and PRISMA totals remain anchored to the preserved import and screening records. For Scopus, the exact archived advanced-search syntax was:

```
TITLE-ABS-KEY (( "truck scheduling" OR "truck appointment system"
OR "yard management"
OR "terminal logistics" OR "agribusiness logistics" )
AND ( queue* OR "resource allocation" OR bottleneck OR dock* )
AND ( optimiz* OR heuristic* OR "reinforcement learning" OR "machine
learning" OR simulation )
AND ( "real-time" OR dynamic OR uncertain* ))
AND PUBYEAR > 2020 AND PUBYEAR < 2027
```

This exact Scopus query returned 22 documents. Deduplication is auditable only at aggregate level in the current archive: 18 duplicates were removed, leaving 42 records for screening. No additional records were excluded on topical grounds at title/abstract screening; the screening loss came from 13 records whose full texts remained inaccessible despite exhaustive retrieval attempts through institutional subscriptions, CAPES, institutional repositories, and public author pages. This availability-based attrition corresponds to 31.0% of the post-deduplication set (13/42). Backward snowballing then added eight studies, yielding 37 studies entering quality assessment and 34 studies passing the quality threshold. To align the final synthesis with the terminal-centered framing of this journal version, three peripheral non-terminal analogues were then removed from the main corpus: two autonomous mining-truck scheduling studies and one expressway charging/mobile-storage truck-scheduling study. The final retained corpus analyzed below therefore contains 31 studies. The pairwise duplicate log and the article-level metadata of the 13 inaccessible full-text records were not preserved in the current workspace.

Backward snowballing inspected the reference lists of the screened papers through Zotero and applied the same inclusion and exclusion rules used in the main search. This step was used to recover antecedent or adjacent studies that fit the review scope but were not returned directly by the database query.

Inclusion required studies to: (i) address dynamic operations, re-optimization, or uncertainty; (ii) focus on terminals, yards, cross-docking systems, port ar-

eas, intermodal freight terminals, or comparable bounded freight-handling facilities with internal service bottlenecks; (iii) propose or evaluate a computational artifact for physical resource allocation; (iv) report some form of validation, such as computational experiments, simulation, benchmarking, or case-based analysis; and (v) be published as peer-reviewed journal or conference papers. Exclusion removed non-technical studies, routing-only papers without internal facility bottlenecks, duplicates, inaccessible full texts, review or conceptual papers, and peripheral non-terminal analogues such as mining-haulage or highway/mobile-charging systems. The temporal filter 2021–2026 was applied in Scopus and Web of Science, but not in IEEE Xplore or snowballing, to preserve earlier antecedent works later screened by the same eligibility rules.

2.2. Quality Assessment

The protocol used a ten-question quality-assessment checklist. Each item was scored as *Yes* = 1, *Partially* = 0.5, and *No* = 0, and studies were retained only if the total score was > 7.5. The checklist asked:

1. Is the objective of the modeling clearly defined?
2. Do the results provide useful evidence for real-world applications of truck orchestration in yards, terminals, or comparable freight-handling facilities?
3. Are the limitations, simplifications, and threats to validity explicitly stated?
4. Does the study test relevant scenarios, events, or operational disruptions?
5. Does the study present validation through experiments, simulation, benchmarking, or computational study?
6. Does the study deliver a clear technical artifact, such as a model, algorithm, architecture, prototype, or system?
7. Does the study explicitly address uncertainty, operational dynamics, and/or real-time re-optimization?
8. Is the proposed method described in sufficient detail for understanding and analysis?
9. Are the decisions, constraints, and objective function clearly presented?
10. Are the logistics focus and the operational context clearly described?

Thirty-seven studies entered quality assessment, three scored ≤ 7.5, and 34 passed the quality threshold. A final terminal-focused refinement then removed three peripheral non-terminal analogues, yielding a retained corpus of 31 studies. Five reviewers used the same screening and extraction form throughout the process. Uncertain inclusion decisions, quality judgments, and later frontier assignments were resolved by joint adjudication so that eligibility and coding criteria remained aligned across the corpus.

2.3. Extraction and Analytical Synthesis

For each retained study, the extraction form captured DOI or ISBN, authors and year, application environment, logistics focus, modeling objective, uncertainty treatment, dynamic feature, technical artifact delivered, validation type, events or disruptions tested, performance metrics reported, limitations or gaps, and emerging requirements. This shared structure supported both descriptive synthesis and the interpretive reading developed later in the paper.

The extraction form fed the final analysis in two stages. First, descriptive coding summarized where the literature concentrates in terms of environment, method family, uncertainty treatment, dynamic mode, disruptions, validation strategy, and reported metrics. Second, interpretive coding translated the extraction sheet into the *validity frontier*. Operational pain points were coded primarily from logistics focus and modeling objectives, whereas structural simplifications were coded from assumptions, limitations, and omitted operational factors. These codes were then read through four dimensions: subsystem integration (isolated $\rightarrow$ integrated), uncertainty richness (narrow $\rightarrow$ richer), response mode (offline $\rightarrow$ adaptive), and validation grounding (solver-only $\rightarrow$ simulation or operational evidence). Each study was positioned qualitatively along these four dimensions using the shared extraction form and joint adjudication among reviewers. For example, the predictive-reactive cross-docking model of Nasiri et al. sits in an intermediate position because it addresses truck-arrival uncertainty and rescheduling but remains grounded mostly in constructed instances, whereas the ETA-informed retail case of Modica et al. lies nearer the frontier because it couples adaptive scheduling with case-based operational evidence [1, 21]. In practical terms, deterministic dock scheduling with isolated resources sits farther from the frontier, whereas ETA-updated or event-stream-driven rescheduling sits nearer. Appendix Appendix A summarizes the current preservation status

of the review audit trail, Appendix Appendix B lists the 31 retained studies, and the complete extraction spreadsheet accompanies the manuscript package as `data_extraction.xls`.

Table 1: Source-level search execution details preserved for retrospective audit.

Source	Execution date	Recoverable retrieval scope	Filters applied	Preserved count
IEEE Xplore	February 27, 2026	Exact advanced-search syntax preserved for this source; direct Boolean query with no year filter.	No temporal filter.	13 results shown retrospectively; 12 imported records preserved in the archived snapshot.
Web of Science	February 27, 2026	Exact advanced-search syntax preserved for this source; `TS=` retrieval with publication restriction.	Publication-year filter 2021–2026.	21 results shown retrospectively; 20 imported records preserved in the archived snapshot.
Scopus	February 27, 2026	Exact advanced-search syntax preserved for this source; `TITLE-ABS-KEY` retrieval with year filter.	Publication-year filter 2021–2026.	22 documents found and imported.
Backward snowballing via Zotero	Same review cycle after database screening	Reference lists of screened papers; no database-specific field syntax applies.	Same inclusion and exclusion rules as the database search.	8 additional records.

3. Findings

3.1. Corpus Profile

After excluding three peripheral non-terminal analogues from the final synthesis, the 31 analyzed studies are concentrated in cross-docking centers (12 studies) and container terminals (11 studies). The remaining works address generic logistics systems with facility-side truck/resource bottlenecks (2 studies), intermodal rail terminals (2 studies), or do not clearly specify the application environment (4 studies). The corpus still includes cold-chain cross-docks, Physical Internet hubs, grocery-retailing unloading operations, and mixed-door configurations, but all retained studies preserve bounded freight-handling bottlenecks comparable to terminal orchestration [7, 16, 21, 22].

3.2. Evidence Maps

To preserve study-level traceability without overloading the page, Figs. 2 and 3 convert the coded extraction sheet into two corpus maps. Figure 2 shows where the literature concentrates across environments and dominant operational pain points, whereas Fig. 3 links dynamic sophistication to validation maturity. In both maps, cell values denote study counts rather than performance scores.

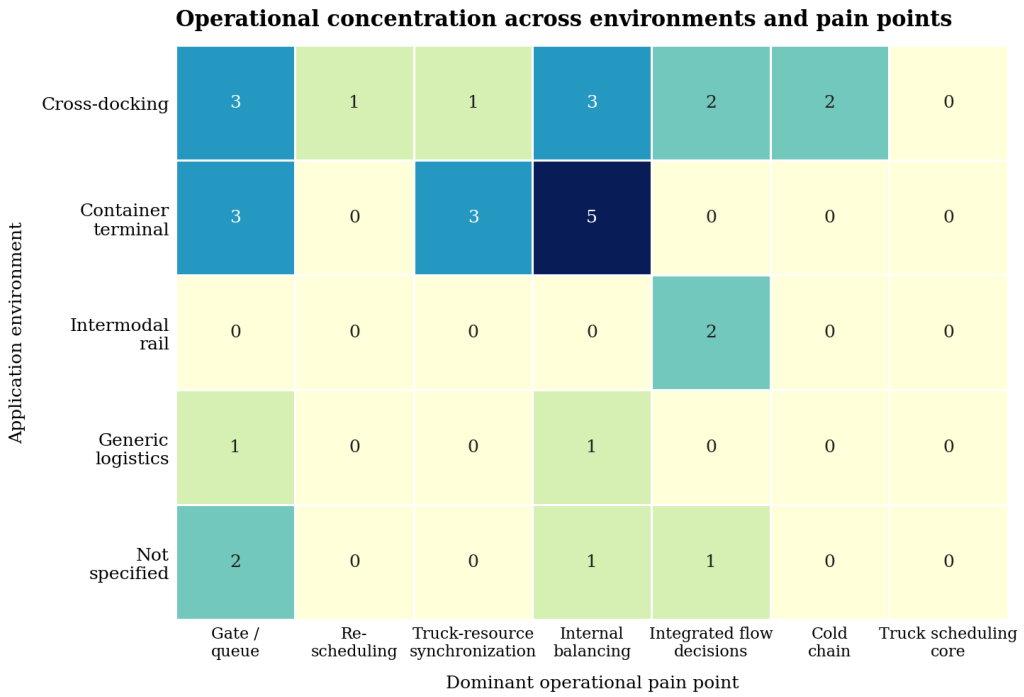

Figure 2: Operational concentration across application environments and dominant operational pain points. Darker cells indicate that more studies concentrate in the corresponding combination.

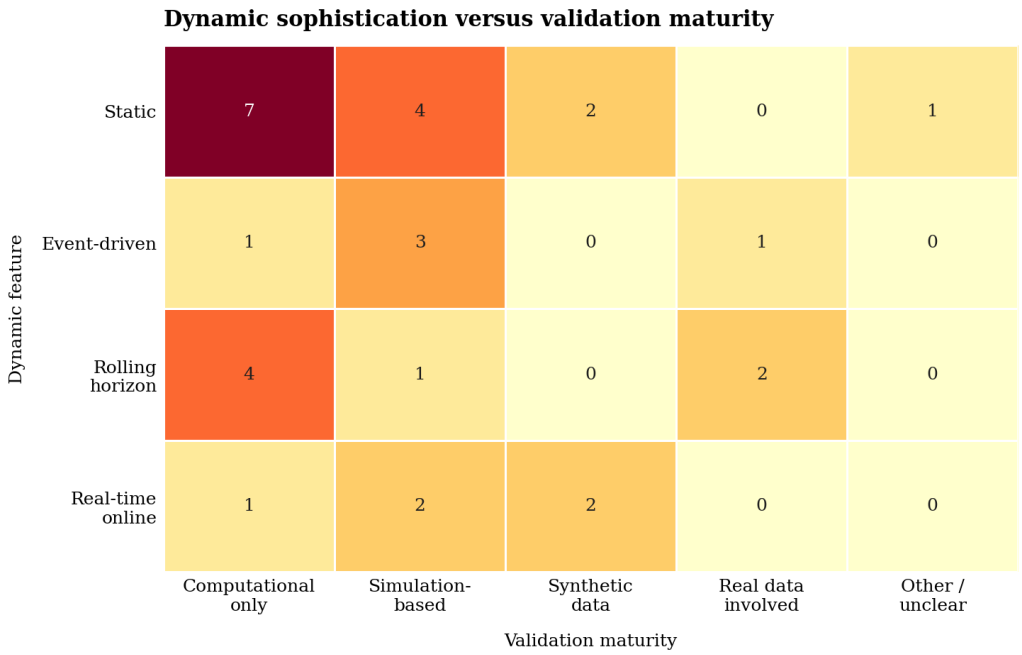

Figure 3: Dynamic sophistication versus validation maturity. Darker cells indicate that more studies occupy the corresponding combination, exposing the gap between adaptive logic and real-data evidence.

The maps make two patterns immediately visible. First, the corpus is concentrated in cross-docking and container-terminal settings, especially around gate congestion, internal balancing, and truck-resource synchronization. Second, adaptive studies still rely mostly on computational experiments and simulation, with little real-data validation. At aggregate level, mathematical programming remains the dominant backbone, with metaheuristics, hybrid optimization or learning artifacts, and simulation-based models forming smaller but recurring streams; explicit disruption testing remains relatively scarce and validation is still predominantly computational or simulation-based [4, 8].

3.3. RQ1: Computational Approaches

Across the 31-study corpus, mathematical programming remains the dominant methodological backbone, appearing in 21 studies (68%), followed by metaheuristics in 15 (48%), hybrid optimization or learning artifacts in 11 (35%), and simulation or agent-based models in 9 (29%). Cross-docking

studies more often emphasize temporary inventory, door synchronization, and tardiness control, whereas container-terminal and yard-oriented studies emphasize gate congestion, yard allocation, crane-truck coordination, and uncertain retrieval from storage [1, 3, 11]. A useful contrast is therefore between MILP-heavy cross-docking formulations and simulation-coupled or agent-based terminal rescheduling, which pursue the same orchestration goal through very different evidence strategies [1, 3, 9]. A smaller stream combines optimization with richer signals or auxiliary learning, but even there the main decision engine usually remains an optimization or simulation core rather than end-to-end learning [8, 11, 23].

3.4. RQ2: Objectives, Dynamics, and Technical Artifacts

Objectives and Metrics. Operational objectives remain strongly dominated by temporal control. Across the reviewed studies, schedule adherence, waiting-time reduction, queue mitigation, and makespan minimization appear more often than explicitly environmental or broader systemic objectives. This pattern is visible across cross-docking, container-terminal, and closely related facility-side settings, where models typically seek to reduce tardiness, synchronize truck arrivals and service, or shorten overall completion time [1, 2, 7, 21–24]. Even when cost is modeled explicitly, it is usually decomposed into penalties associated with lateness, queuing, or appointment changes, which makes cost a monetary translation of temporal efficiency [5, 6, 25, 26]. Objective-metric alignment is also imperfect: some studies optimize waiting-time, stability, or workload balancing while foregrounding downstream indicators such as CO_2 reduction, throughput, or utilization in the interpretation of results [2, 13, 14].

Dynamic Features and Artifacts. At the artifact level, the corpus is transitional rather than fully real-time. Many adaptive studies adopt predictive-reactive or rolling-horizon logic, with ETA updates, appointment reconfirmation, or event-triggered revisions, but continuous online control remains rare [1, 2, 5, 8, 9, 12, 21]. A smaller stream combines optimization with richer signals or auxiliary learning, but even there the main decision engine usually remains an optimization or simulation core rather than end-to-end learning [8, 11, 23]. Environmental performance therefore appears as an emerging but secondary analytical axis, often embedded in congestion reduction, idling control, or appointment rescheduling rather than treated as an independent optimization objective [5, 14, 25].

3.5. RQ3: Validation Gaps and Practical Applicability

Validation maturity remains limited. Across the 31-study corpus, 29 studies (94%) rely on computational experiments, 13 (42%) include simulation, and only three (10%) incorporate any real data. Solver-oriented validation dominates static or quasi-static settings, while simulation-backed evaluation is the main mechanism used to test adaptive behavior under stochastic arrivals, resource interactions, or event-driven control [2–4, 6, 7, 9, 22, 23, 26, 27]. Synthetic data are used chiefly for benchmark-instance generation, perturbation or scenario generation, and pseudo-operational validation, which strengthens algorithmic comparison but does not close the operational-grounding gap [13, 14, 21].

The most recurrent practical limitations follow a recognizable pattern. Many studies isolate one subsystem while leaving upstream and downstream interactions outside the model, treat trucks, cargo, or handling resources as more homogeneous than they are in practice, assume oversized or effectively unlimited temporary buffers, simplify terminal topology and internal travel, or rely on deterministic parameters and highly visible system states [4, 6, 11, 13, 14, 22]. These simplifications weaken external validity because they suppress spillback, cross-subsystem propagation, and human or infrastructure frictions. In practice, many published orchestration policies remain analytically strong but structurally under-stressed relative to real terminal operations.

4. Discussion

The field already offers a broad toolbox for well-bounded scheduling and allocation problems, but deployable orchestration remains less mature. Terminal operations require synchronization across multiple resources, changing truck arrivals, and imperfect information streams [1–3], yet many published models still assume simplified disruptions, stable capacities, or centralized information availability [4, 12, 15, 16].

4.1. Analytical Matrix: Pain Points Versus Validity Frontier

The corpus can be read through two complementary dimensions: recurring logistics pains and the structural simplifications used to keep NP-hard models tractable. Table 2 combines these dimensions and summarizes the main interpretive lens of this review.

Under this lens, many solutions that appear optimal at the algorithmic level are optimal only inside a sanitized operating environment. Recurring assumptions such as homogeneous loads, negligible transfer times, oversized temporary buffers, or partial uncertainty models create a gap between solver success and operational credibility [6, 13, 16]. What moves a study nearer the proposed boundary is not simply the presence of dynamic logic, but the combination of broader uncertainty, adaptive updates, subsystem coupling, and validation beyond solver benchmarks.

The matrix also clarifies why methodological sophistication alone is not enough. ETA-informed rescheduling, event-triggered control, and hybrid optimization or learning architectures are meaningful advances, but most are still validated under narrower disturbance spaces and cleaner information environments than real terminals face [4, 5, 7, 8, 11, 13, 21]. In that sense, the main dividing line in the literature is less exact versus heuristic optimization and more stylized scheduling performance versus orchestration claims that remain credible when gate, yard, dock, labor, and handling interactions are restored.

This reading also helps position secondary themes such as sustainability and ML. Environmental gains are usually framed as beneficial byproducts of temporal optimization rather than as independently validated trade-offs, while ML typically augments optimization by tuning solvers or exploiting richer event streams instead of replacing the decision core [5, 8, 11, 14, 23, 25]. The field is therefore moving toward data-coupled orchestration, but its clearest open need remains operationally grounded validation that tests whether algorithmic gains survive cross-stakeholder data barriers, subsystem coupling, and live execution frictions.

Table 2: Analytical matrix linking operational pain points to dominant structural simplifications and external-validity risks. Read each row from left to right: pain point, operational focus, and the simplifying assumption that most threatens external validity.

Pain point	Operational focus	Dominant simplification and validity risk
Gate congestion and external queues	Protect appointments, reduce idling, and prevent entrance overload.	Gate operations are often modeled apart from the yard and quay, so queue relief may disappear once internal congestion propagates.
Disruption-driven rescheduling	Repair obsolete plans after delays, uncertainty release, or workload shifts.	ETA-based disturbance sets are usually narrow, so schedule stability can collapse under coupled events or stale data.
Truck-resource synchronization	Prevent truck queues and crane, forklift, or labor idleness.	Identical resources, ignored transfer times, and simplified topology can overstate the level of synchronization that is actually attainable.
Internal balancing in yard or cross-dock	Prevent overflow, re-handlings, uneven temporary inventory, and low yard utilization.	Oversized buffers, homogeneous loads, or restricted equipment moves mask spillback, space scarcity, and shutdown risk.
Integrated service-window decisions	Coordinate dock sequencing with routing, transfer distance, or multimodal timing.	Deterministic transfer times and stylized waiting assumptions weaken claims of integrated optimality under heterogeneous operations.
Cold-chain scheduling	Limit tardiness, spoilage, and stockout under perishability-sensitive service.	Selective uncertainty and stylized degradation logic can overestimate robustness when spoilage and service variability interact jointly.

5. Conclusion and Limitations

This review maps a field that is rich in scheduling methods but still uneven in subsystem integration, disturbance modeling, and empirical grounding. Across the 31-study terminal-focused corpus, terminal-orchestration research is strongest when solving well-bounded temporal coordination problems and weaker when asked to demonstrate that those gains persist under richer operational complexity. The main contribution of the paper is the exposure of a *validity frontier* for terminal-orchestration research, showing where current approaches remain analytically strong and where they still have narrow operational grounding.

For future work, the evidence points less to new solvers per se than to stronger testbeds and integration assets: shared benchmarks, open synthetic generators tied to realistic signals, digital-twin loops, and studies that connect gate, yard, dock, and handling resources under richer disturbances. For research that claims adaptive or real-time behavior, progress now depends primarily on validation that goes beyond stylized instances and single-site pseudo-operational settings.

Methodologically, the review has limitations. Most importantly, 13 of the 42 post-deduplication records (31.0%) were excluded because their full texts remained inaccessible even after exhaustive searches through institutional subscriptions, CAPES, institutional repositories, and public author pages. Because the current workspace preserves only the aggregate count, not the definitive article-level list of those records, no source-, topic-, or year-specific sensitivity analysis on that availability loss can yet be performed. The quality cutoff of > 7.5 is preserved as a rule, but the study-by-study score matrix and score distribution are not, so threshold-robustness analyses cannot be reproduced retrospectively from the current archive. Because the query was intentionally targeted to terminal, method, and performance terms, relevant studies using different vocabulary may also have been missed. In addition, although the screening and extraction process involved five reviewers using a shared form and joint resolution of disagreements, no formal inter-rater agreement statistic was computed.

Appendix A. Audit Trail and Preservation Status

The most consequential limitation is availability-based attrition. The 13 inaccessible full-text records represent 31.0% of the post-deduplication set

(13/42) and 41.9% of the size of the final retained corpus (13/31). This is not a topical exclusion but a loss driven by document access. The current workspace therefore supports a transparent report of the magnitude of this bias, but not a definitive appendix listing those 13 studies with article-level metadata. Recovering the original Parsifal screening log remains necessary for a journal-grade audit appendix on full-text loss and for any formal sensitivity analysis around that attrition.

Table A.3: Preservation status of the main review artifacts in the current manuscript package.

Artifact	Status	Current evidence in the package
Core Boolean query	Preserved	Reported in Section 2.1 and companion audit notes.
Source list, execution date, and imported counts	Preserved	Table 1 records the three databases, the search date, the source-level counts, and the snowballing additions.
Database-specific field tags typed in each interface	Preserved	The exact IEEE Xplore, Web of Science, and Scopus search strings are now archived, although only the aggregate deduplication and screening logs were preserved.
Duplicate-pair log	Not preserved	The package preserves the aggregate result of deduplication (18 removals), but not the pairwise duplicate register.
Final retained-corpus list	Preserved	Appendix Appendix B enumerates the 31 included studies in the terminal-focused synthesis.
Full extraction spreadsheet	Preserved	The manuscript package includes `data_extraction.xls`.
Article-level list of the 13 inaccessible full-text records	Not authoritatively preserved	Only the aggregate count is preserved; current Zotero collections allow partial reconstruction of non-retained records, but they do not separate full-text loss cleanly from topical exclusions, duplicate versions, or later replacements.
Item-level quality responses, per-study totals, and score distribution	Not preserved	The checklist, answer scale, threshold rule, and retained corpus are preserved, but the complete score matrix is not.

Appendix B. Included Studies in the Retained Corpus

Table B.4: Studies retained in the final 31-paper corpus.

Authors (year)	Study title	DOI/ISBN
Jin et al. (2021)	A Dynamic Joint Real-Time Scheduling Strategy of Truck and Yard Crane in Container Terminals	10.1109/ICNSC52481.2021.9702168
Li and Li (2010)	Modeling and Simulation of Container Terminal Logistics Systems Using Harvard Architecture and Agent-Based Computing	10.1109/WSC.2010.5679030
Li et al. (2009)	Modeling and Simulation of Yard Trailer Dispatching at Container Terminals	10.1109/ICAL.2009.5262988
Liu and Liu (2016)	The Hybrid Intelligence Swam Algorithm for Berth-Quay Cranes and Trucks Scheduling Optimization Problem	10.1109/ICCI-CC.2016.7862049
Supeno et al. (2015)	Optimum Scheduling System for Container Truck Based on Genetic Algorithm on Spin(Simulator Pelabuhan Indonesia)	10.1109/IDM.2015.7516352
Zouhaier and Ben Said (2016)	An Application Oriented Multi-Agent Based Approach to Dynamic Truck Scheduling at Cross-Dock	10.1109/PDCAT.2016.058
da Silva et al. (2023)	Integration of Machine Learning and Simulation for Dynamic Rescheduling in Truck Appointment Systems	10.1016/j.simpat.2023.102747
Essghaier et al. (2023)	Fuzzy Multi-Objective Truck Scheduling in Multi-Modal Rail-Road Physical Internet Hubs	10.1016/j.cie.2023.109404
Khodashenas et al. (2026)	The Cost and Time Objectives Minimization in Cross-Dock Truck Scheduling of Perishable Goods Considering Uncertainty	10.5829/ije.2026.39.02b.16
Luan et al. (2025)	An Efficient Real-Time Railway Container Yard Management Method Based on Partial Decoupling	10.1109/TASE.2025.3559241
Movassaghi and Darestani (2022)	Multiple Cross-Docks Scheduling with Multiple Doors Using Fuzzy Approach and Metaheuristic Algorithms	10.1007/s40305-021-00362-9
Nasiri et al. (2022)	A Predictive-Reactive Cross-Dock Rescheduling System under Truck Arrival Uncertainty	10.1016/j.eswa.2021.115986
Wang et al. (2023)	Flexible Storage Yard Management in Container Terminals under Uncertainty	10.1016/j.cie.2023.109753
Xu et al. (2021)	Optimization for a Multi-Constraint Truck Appointment System Considering Morning and Evening Peak Congestion	10.3390/su13031181

Continued on next page

Authors (year)	Study title	DOI/ISBN
Bouyahia et al. (2025)	A Novel Truck Appointment System for Container Terminals	10.3390/su17135740
Cota et al. (2022)	Integrating Vehicle Scheduling and Open Routing Decisions in a Cross-Docking Center with Multiple Docks	10.1016/j.cie.2021.107869
Fonseca et al. (2025)	Stability Approach to CDC Truck Scheduling Problem under Uncertainty	10.1007/s11590-024-02137-6
Kappatos et al. (2025)	Truck Service Optimisation in Port Areas by Comparing Genetic Algorithm and Cuckoo Search Algorithm	10.2478/ttj-2025-0030
Movassaghi and Darestani (2021)	Cross-Docks Scheduling with Multiple Doors Using Fuzzy Approach	10.48295/ET.2020.79.3
Sha et al. (2021)	Simulation Model to Determine Ratios between Quay, Yard and Intra-Terminal Transfer Equipment in an Integrated Container Handling System	10.24006/JILT.2021.19.1.001
Torbali and Alpan (2023)	A Multi-Agent-Based Real-Time Truck Scheduling Model for Cross-Docking Problems with Single Inbound and Outbound Doors	10.1016/j.sca.2023.100028
Zheng et al. (2026)	Truck Scheduling Optimization at a Cold Chain Cross-Docking Terminal Considering Uncertainties and the Door-Mixed Service Mode	10.1016/j.eswa.2025.129849
Abbas et al. (2018)	A Constrained Fuzzy Knowledge-Based System for the Management of Container Yard Operations	10.1007/s40815-018-0448-9
Cekała et al. (2015)	Truck Loading Schedule Optimization Using Genetic Algorithm for Yard Management	10.1007/978-3-319-15702-3_52
Vahdani and Shahramfard (2019)	A Truck Scheduling Problem at a Cross-Docking Facility with Mixed Service Mode Dock Doors	10.1108/EC-08-2018-0355
Xu et al. (2022)	Dynamic Appointment Rescheduling of Trucks under Uncertainty of Arrival Time	10.3390/jmse10050695
Banyai and Trojahn (2025)	Adaptive Real-Time Planning of Trailer Assignments in High-Throughput Cross-Docking Terminals	10.3390/a18110679
Eti et al. (2025)	Multi-Door Cross-Dock Scheduling under Flexible Doors Mode and Material Handling Resource Restrictions	10.1016/j.cor.2025.107183
Modica et al. (2024)	Integrating Arrival Time Estimation in Truck Scheduling: An Explorative Study in Grocery Retailing	10.1080/21693277.2024.2425678

Continued on next page

Authors (year)	Study title	DOI/ISBN
Noh et al. (2025)	Inbound Truck Scheduling for Workload Balancing in Cross-Docking Terminals	10.3390/math 13152533
Rostami et al. (2022)	Modelling Cross-Docking in a Three-Level Supply Chain with Stochastic Service and Queuing System: MOWFA Algorithm	10.3390/a15080 265

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This article is a systematic literature review and does not report a new experimental dataset. The manuscript package includes the retained-corpus extraction spreadsheet (`data_extraction.xls`), companion audit notes, and the appendix-level list of included studies. The current package does not yet include the definitive duplicate-pair log, the article-level list of the 13 inaccessible full-text records, or the item-level quality-score matrix; those artifacts require recovery of the original Parsifal workspace used during screening.

References

[1] M. M. Nasiri, N. Ahmadi, D. Konur, A. Rahbari, A predictive-reactive cross-dock rescheduling system under truck arrival uncertainty, Expert Syst. Appl. 188 (2022) 115986. doi:10.1016/j.eswa.2021.115986.

[2] G. B. Fonseca, T. H. Nogueira, M. G. Ravetti, Stability approach to CDC truck scheduling problem under uncertainty, Optim. Lett. 19 (4) (2025) 811–832. doi:10.1007/s11590-024-02137-6.

[3] A. Abbas, A. Al-Bazi, V. Palade, A constrained fuzzy knowledge-based system for the management of container yard operations, Int. J. Fuzzy Syst. 20 (4) (2018) 1205–1223. doi:10.1007/s40815-018-0448-9.

[4] P. M. Cota, T. H. Nogueira, A. A. Juan, M. G. Ravetti, Integrating vehicle scheduling and open routing decisions in a cross-docking center with multiple docks, Comput. Ind. Eng. 164 (2022) 107869. doi:10.1016/j.cie.2021.107869.

[5] B. Xu, X. Liu, J. Li, Y. Yang, J. Wu, Y. Shen, Y. Zhou, Dynamic appointment rescheduling of trucks under uncertainty of arrival time, J. Mar. Sci. Eng. 10 (5) (2022). doi:10.3390/jmse10050695.

[6] B. Vahdani, S. Shahramfard, A truck scheduling problem at a cross-docking facility with mixed service mode dock doors, Eng. Comput. (Swans. Wales) 36 (6) (2019) 1977–2009. doi:10.1108/EC-08-2018-0355.

[7] N. A. Eti, W. G. Ferrell, N. Huynh, Multi-door cross-dock scheduling under flexible doors mode and material handling resource restrictions, Comput. Oper. Res. 183 (2025) 107183. doi:10.1016/j.cor.2025.107183.

[8] M. R. F. da Silva, I. R. S. Agostino, E. M. Frazzon, Integration of machine learning and simulation for dynamic rescheduling in truck appointment systems, Simul. Modell. Pract. Theory 125 (2023) 102747. doi:10.1016/j.simpat.2023.102747.

[9] B. Torbali, G. Alpan, A multi-agent-based real-time truck scheduling model for cross-docking problems with single inbound and outbound doors, Supply Chain Anal. 3 (2023) 100028. doi:10.1016/j.sca.2023.100028.

[10] X. Jin, X. Wang, W. Wu, Z. Rong, Q. Zhang, A dynamic joint real-time scheduling strategy of truck and yard crane in container terminals, in: 2021 IEEE International Conference on Networking, Sensing and Control (ICNSC), Vol. 1, IEEE, Xiamen, China, 2021, pp. 1–6. doi:10.1109/ICNSC52481.2021.9702168.

[11] Y. Luan, Q.-S. Jia, Y. Xing, Z. Li, T. Wang, An efficient real-time railway container yard management method based on partial decoupling, IEEE Trans. Autom. Sci. Eng. 22 (2025) 14183–14200. doi:10.1109/TASE.2025.3559241.

[12] T. Bányai, S. Trojahn, Adaptive real-time planning of trailer assignments in high-throughput cross-docking terminals, Algorithms 18 (11) (2025) 679. doi:10.3390/a18110679.

[13] Y. Noh, S. Lee, J. Hong, J. Kim, S. W. Cho, Inbound truck scheduling for workload balancing in cross-docking terminals, Mathematics 13 (15) (2025) 2533. doi:10.3390/math13152533.

[14] F. Bouyahia, S. Belaqziz, Y. Meliani, S. L. Elhaq, J. Boukachour, A novel truck appointment system for container terminals, Sustainability 17 (13) (2025) 5740. doi:10.3390/su17135740.

[15] P. Rostami, S. Avakh Darestani, M. Movassaghi, Modelling cross-docking in a three-level supply chain with stochastic service and queuing system: MOWFA algorithm, Algorithms 15 (8) (2022) 265. doi:10.3390/a15080265.

[16] F. Essghaier, T. Chargui, T. Hsu, A. Bekrar, H. Allaoui, D. Trentesaux, G. Goncalves, Fuzzy multi-objective truck scheduling in multi-modal rail-road physical internet hubs, Comput. Ind. Eng. 182 (2023) 109404. doi:10.1016/j.cie.2023.109404.

[17] N. Boysen, M. Fliedner, Cross dock scheduling: Classification, literature review and research agenda, Omega 38 (6) (2010) 413–422. doi:10.1016/j.omega.2009.10.008.

[18] T. F. Abdelmagid, M. Dessouky, S. Ricci, N. A. Harraz, Truck appointment systems in container terminals: A literature review, Annals of Operations Research 318 (2022) 2075–2124. doi:10.1007/s10479-022-05083-z.

[19] M. A. Naeem, M. Alharthi, H. Khayyam, A. K. Rathore, N. Akhter, M. Shahzad, O. W. Samuel, A comprehensive review of integrated scheduling methods in automated container terminals, Inventions 8 (1) (2023) 35. doi:10.3390/inventions8010035.

[20] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness,

L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, D. Moher, The PRISMA 2020 statement: An updated guideline for reporting systematic reviews, BMJ (2021) n71doi:10.1136/bmj.n71.

[21] T. Modica, E. Tappia, C. Colicchia, M. Melacini, Integrating arrival time estimation in truck scheduling: An explorative study in grocery retailing, Prod. Manuf. Res. 12 (1) (2024) 2425678. doi:10.1080/21693277.2024.2425678.

[22] F. Zheng, Y. Yi, M. Liu, H. Qiu, Truck scheduling optimization at a cold chain cross-docking terminal considering uncertainties and the door-mixed service mode, Expert Syst. Appl. 298 (C) (2026) 129849. doi:10.1016/j.eswa.2025.129849.

[23] T. H. Nogueira, F. P. Coutinho, M. G. M. Peixoto, E. G. Carrano, M. G. Ravetti, Evolutionary algorithm framework for optimizing truck scheduling in multi-dock truck cross-docking centers, Evol. Intell. 18 (1) (2025) 2. doi:10.1007/s12065-024-00992-x.

[24] T. Cekała, Z. Telec, B. Trawiński, Truck loading schedule optimization using genetic algorithm for yard management, in: Lect. Notes Comput. Sci., Vol. 9011, Springer Verlag, 2015, pp. 536–548. doi:10.1007/978-3-319-15702-3_52.

[25] B. Xu, X. Liu, Y. Yang, J. Li, O. Postolache, Optimization for a multi-constraint truck appointment system considering morning and evening peak congestion, Sustainability 13 (3) (2021) 1181. doi:10.3390/su13031181.

[26] W. Wang, S. Lin, L. Zhen, Flexible storage yard management in container terminals under uncertainty, Comput. Ind. Eng. 186 (2023) 109753. doi:10.1016/j.cie.2023.109753.

[27] B. Li, W.-f. Li, Modeling and simulation of container terminal logistics systems using harvard architecture and agent-based computing, in: Proceedings of the 2010 Winter Simulation Conference, IEEE, Baltimore, MD, USA, 2010, pp. 3396–3410. doi:10.1109/WSC.2010.5679030.